SUBJECT OUTLINE

Subject code and name
MLN601 Machine Learning

SECTION 1 – GENERAL INFORMATION

1.1 Subject level and weighting

Subject level	Subject credit points
600	10

1.2 Student Workload

Duration	Facilitated study h/p/w	Personal study h/p/w	Total study h/p/w
12-week duration	3	7	10
6-week duration	6	14	20

1.3 Delivery mode

☒ Face to face

☒ Online

☐ Intensive mode (e.g. residential, or summer school, weekend workshop – specify details below)

☐ Work-integrated learning activity

☒ Hybrid

☒ Full-time

☒ Part-time

☐ Other (please specify)

1.4 Pre-requisites and co-requisite

☒ Yes ☐ No

If YES, prove details of the pre-requisite or co-requisite requirements below.

MFA501 – Mathematical Foundations of AI

1.5 Specialist facilities and/or equipment requirements

☒ Yes ☐ No

If YES, provide details of specialist facilities and/or equipment below.

Computer (with Jupyter and Python installed)

SECTION 2 – ACADEMIC DETAILS

2.1 Subject Descriptor

This subject is designed to provide a graduate-level student an in-depth understanding of the methodologies, technologies, and algorithms currently used in machine learning. Students will be introduced to the theory behind a range of machine learning tools and practice applying the tools to different applications. It explores topics such as classification, linear models, supervised and unsupervised learning in conjunction with applied and theoretical machine learning concepts. Students complete practical activities using machine learning algorithms and methods.

2.2 Subject Learning Outcomes

Subject Learning Outcomes
a) Evaluate and compare the key concepts in machine learning.
b) Apply learning algorithms to perform machine learning tasks.
c) Implement practical machine learning: data pre-processing, analysis, model selection, and interpret the results.
d) Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders.

2.3 Assessment tasks

a. 12-week duration

No.	Type and Description	Assessment due	Weighting (%)	Subject Learning Outcome(s) assessed
1	**Title:** Regression Analysis **Delivery type:** Individual **Outcome:** Notebook with mark-up **Volume of Assessment:** Source code and 1000 words (+/- 10%)	Module 4 (Week 4)	20%	b, c, d
2	**Title:** Classification **Delivery type:** Individual **Outcome:** Notebook with mark-up and presentation	Module 8 (Week 8)	40%	b, c, d

	Volume of Assessment: 7-10 min presentation and source code, 1500 words (+/- 10%).			
3	**Title:** Machine learning project **Delivery type:** Individual **Outcome:** Notebook and model selection **Volume of Assessment:** up to 2000 words	Module 12 (Week 12)	40%	a, b, c, d

b. 6-week duration

No.	Type and Description	Assessment due	Weighting (%)	Subject Learning Outcome(s) assessed
1	**Title:** Regression Analysis **Delivery type:** Individual **Outcome:** Notebook with mark-up **Volume of Assessment:** Source code and 1000 words (+/- 10%)	Module 4 (Week 2)	20%	b, c, d
2	**Title:** Classification **Delivery type:** Individual **Outcome:** Notebook with mark-up and presentation **Volume of Assessment:** 7-10 min presentation and source code, 1500 words (+/- 10%).	Module 8 (Week 4)	40%	b, c, d
3	**Title:** Machine learning project **Delivery type:** Individual **Outcome:** Notebook and model selection **Volume of Assessment:** up to 2000 words	Module 12 (Week 6)	40%	a, b, c, d

2.4 Prescribed texts

These are core texts that are fundamental to meeting subject outcomes. Students may choose to purchase a copy. A limited number of these texts will be available for loan from the campus library.

N/A

2.5 Suggested readings and resources

These are supplementary readings and resources that support subject knowledge. These may be available from the campus library. Refer to *MLN601 Machine Learning* for the list of readings and resources.

Policies and Procedures, and related application forms, are available via the Torrens University Australia website.

2.6 Course Learning Outcomes

Refer to **Student Hub** to access the Course Learning Outcomes.